



MATHEMATICS (EXTENSION 1)

2014 HSC Course Assessment Task 2

Wednesday March 12, 2014

General instructions

- Working time – 55 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets + answer sheet used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer sheet provided (numbered as page 5)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M3A – Mr Zuber

☐ 12M4A – Ms Ziaziaris

☐ 12M3B – Mr Berry

☐ 12M4B – Mr Lam

☐ 12M3C – Mr Lowe

☐ 12M4C – Mr Ireland

Marker's use only.

QUESTION	1-4	5	6	7	8	9	Total	%
MARKS	$\overline{4}$	$\overline{6}$	$\overline{10}$	$\overline{11}$	$\overline{10}$	$\overline{9}$	$\overline{50}$	

Section I: Objective response

Mark your answers on the multiple choice sheet provided.

Marks

1. How many real solutions are there to the equation $e^{2x} + e^x - 6 = 0$? **1**

(A) 0

(C) 2

(B) 1

(D) None of the above

2. Which of the following represent the solution(s) to the equation **1**

$$\ln(x - 2) + \ln(x - 1) = \ln(x + 2)$$

(A) $x = 0, x = 4.$

(C) $x = 0$ only

(B) $x = 4$ only

(D) No real solutions

3. Which of the following represents the primitive of π^x , excluding the constant of integration? **1**

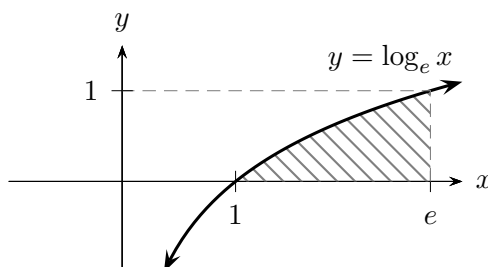
(A) x^π

(B) π^x

(C) $\pi^x \ln \pi$

(D) $\frac{\pi^x}{\ln \pi}$

4. Which of the following represents the correct integral to evaluate the volume generated when the area beneath the curve $y = \log_e x$ between $x = 1$ and $x = e$ is rotated about the y axis? **1**



(A) $\pi \int_0^1 e^{2y} dy$

(C) $\pi \int_1^e (\log_e x)^2 dx$

(B) $\pi e^2 - \pi \int_0^1 e^{2y} dy$

(D) $\pi e^2 - \pi \int_1^e (\log_e x)^2 dx$

End of Section I.
Examination continues overleaf.

Question 5 (6 Marks)

Commence a NEW page.

Marks

- (a) Find the set of values of x for which the limiting sum exists for this series: **2**

$$1 + \ln x + (\ln x)^2 + (\ln x)^3 + \dots$$

- (b) i. Copy and fill in the table of values for $y = \log_e x - 1$. **1**

x	3	4	5	6	7
y					

- ii. By using Simpson's Rule with five function values, find the approximate volume when the curve $y = f(x)$ is rotated about the x axis between $x = 3$ to $x = 7$, correct to 4 decimal places. **3**

Question 6 (10 Marks)

Commence a NEW page.

Marks

- (a) Find $\lim_{x \rightarrow 0} \frac{\sin 9x}{\tan 4x}$. **2**

- (b) i. State the period and amplitude of $y = 2 - \cos \frac{1}{2}x$. **2**

- ii. Hence, sketch the graph of $y = 2 - \cos \frac{1}{2}x$ where $-\pi \leq x \leq \pi$. **2**

- (c) i. Sketch the graph of $y = \tan x$ for $-\pi \leq x \leq \pi$. **2**

- ii. Hence on the same set of axes, sketch the graph of $y = \cot x$. **2**

Question 7 (11 Marks)

Commence a NEW page.

Marks

- (a) Find the derivative of $y = \log_e \left(\sqrt{\frac{1-x^2}{1+x^2}} \right)$. **3**

- (b) i. Differentiate $\frac{2x^2 + 1}{3x^2 - 4}$ with respect to x . **2**

- ii. Hence evaluate $\int \frac{x}{(3x^2 - 4)^2} dx$. **2**

- (c) i. Show that $\frac{x^3}{x+1} = x^2 - x + 1 - \frac{1}{x+1}$. **2**

- ii. Hence evaluate $\int \frac{x^3 - 2}{x+1} dx$. **2**

Question 8 (10 Marks)

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MarksFor the curve $y = xe^x + e^x$,

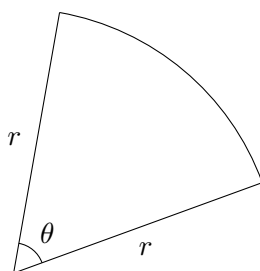
- (a) Find the stationary point on the curve and determine its nature. **3**
- (b) Find any points of inflexion. **3**
- (c) Find where the curve cuts the coordinate axes. **2**
- (d) Hence sketch the curve $y = e^x + xe^x$. **2**

Question 9 (9 Marks)

Commence a NEW page.

Marks

- (a) Prove that $4 \times 2^n + 3^{3n}$ is divisible by 5 for all integers n , $n \geq 1$. **3**
- (b) The diagram shows a sector of a circle with radius r cm. the angle at the centre is θ radians, and the area is 32 cm^2 .



- i. Find an expression for r in terms of θ . **1**
- ii. Show that the perimeter P of the sector is given by **2**

$$P = \frac{8(2 + \theta)}{\sqrt{\theta}}$$

- iii. Find the minimum perimeter and the value of θ for which this occurs. **3**

End of paper.

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

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1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} + C, \quad n \neq -1; \quad x \neq 0 \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x + C, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} + C, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax + C, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax + C, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax + C, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax + C, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} + C, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} + C, \quad a > 0, -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right) + C, \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right) + C$$

NOTE: $\ln x = \log_e x$, $x > 0$

Suggested Solutions

ii. (3 marks)

Section I

1. (B) 2. (B) 3. (D) 4. (B)

Section II

Question 5 (Lam)

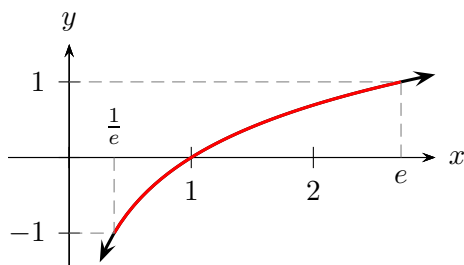
(a) (2 marks)

$$1 + \ln x + (\ln x)^2 + (\ln x)^3 + \dots$$

$$a = 1 \quad r = \ln x$$

Limiting sum exists when $|r| < 1$, i.e.

$$|\ln x| < 1$$

From the graph of $y = \ln x$,the part of $\ln x$ that lies between -1 and 1 occurs when $\frac{1}{e} < x < e$.

(b) i. (1 mark)

- Award full marks for either 3-4 decimal places required for full marks.

x	3	4	5	6	7
y	$\ln 3 - 1$ 0.0986	$\ln 4 - 1$ 0.3863	$\ln 5 - 1$ 0.6094	$\ln 6 - 1$ 0.7918	$\ln 7 - 1$ 0.9459

$$V = \pi \int_a^b y^2 dx$$

$$\approx \pi \times \overbrace{\frac{h}{3} \left((y_0)^2 + 4 \left(\sum (y_{\text{odd}})^2 \right) + 2 \left(\sum (y_{\text{even}})^2 \right) + (y_\ell)^2 \right)}^{\approx \int_a^b y^2 dx}$$

$$V \approx \pi \times \frac{h}{3} \left((\ln 3 - 1)^2 + 4((\ln 4 - 1)^2 + (\ln 6 - 1)^2) + 2((\ln 5 - 1)^2 + (\ln 7 - 1)^2) \right)$$

$$= 4.97599 \dots \approx 4.9760$$

Question 6 (Ziaziaris)

(a) (2 marks)

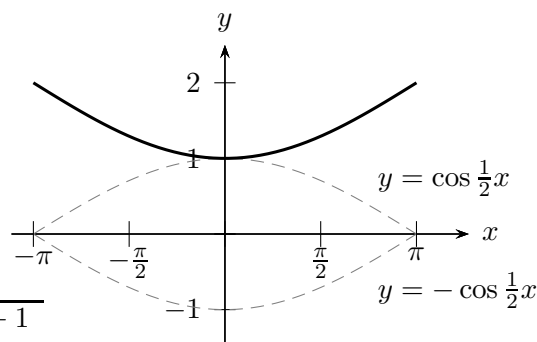
$$\lim_{x \rightarrow 0} \frac{\sin 9x}{\tan 4x} = \lim_{x \rightarrow 0} \left(\frac{\sin 9x}{9x} \times \frac{4x}{\tan 4x} \right) \times \frac{9}{4}$$

$$= \frac{9}{4}$$

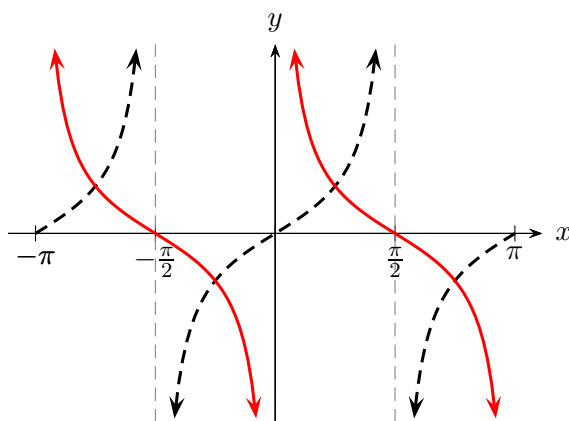
(b) i. (2 marks) $y = 2 - \cos \frac{1}{2}x$

- $T = 4\pi$
- $a = 1$

ii. (2 marks)



(c) (4 marks)

(Dashed line: $y = \tan x$)

(c) i. (2 marks)

$$\begin{aligned}\frac{x^3}{x+1} &= \frac{x^3+1}{x+1} - \frac{1}{x+1} \\ &= \frac{(x+1)(x^2-x+1)}{x+1} - \frac{1}{x+1} \\ &= x^2 - x + 1 - \frac{1}{x+1}\end{aligned}$$

ii. (2 marks)

$$\begin{aligned}\int \frac{x^3-2}{x+1} dx &= \int \left(\frac{x^3}{x+1} - \frac{2}{x+1} \right) dx \\ &= \int \left(x^2 - x + 1 - \frac{3}{x+1} \right) dx \\ &= \frac{1}{3}x^3 - \frac{1}{2}x^2 + x - 3\ln(x+1) + C\end{aligned}$$

Question 7 (Zuber)

(a) (3 marks)

$$\begin{aligned}y &= \ln \sqrt{\frac{1-x^2}{1+x^2}} \\ &= \frac{1}{2} \ln(1-x^2) - \frac{1}{2} \ln(1+x^2)\end{aligned}$$

Differentiating,

$$\begin{aligned}\frac{dy}{dx} &= \frac{-\cancel{2}x}{\cancel{2}(1-x^2)} - \frac{\cancel{2}x}{\cancel{2}(1+x^2)} \\ &= -\frac{x}{1-x^2} - \frac{x}{1+x^2}\end{aligned}$$

(b) i. (2 marks)

$$\begin{aligned}y &= \frac{2x^2+1}{3x^2-4} \\ \begin{cases} u = 2x^2+1 & v = 3x^2-4 \\ u' = 4x & v' = 6x \end{cases} \\ \frac{dy}{dx} &= \frac{4x(3x^2-4) - 6x(2x^2+1)}{(3x^2-4)^2} \\ &= \frac{\cancel{12}x^3 - 16x - \cancel{12}x^3 - 6x}{(3x^2-4)^2} \\ &= -\frac{22x}{(3x^2-4)^2}\end{aligned}$$

ii. (2 marks)

$$\begin{aligned}\int \frac{x}{(3x^2-4)^2} dx &= -\frac{1}{22} \int \frac{-22x}{(3x^2-4)^2} dx \\ &= -\frac{1}{22} \left(\frac{2x^2+1}{3x^2-4} \right) + C\end{aligned}$$

Question 8 (Ireland)

(a) (3 marks)

$$\begin{aligned}y &= xe^x + e^x = e^x(x+1) \\ u &= e^x & v &= x+1 \\ u' &= e^x & v' &= 1 \\ \frac{dy}{dx} &= e^x + e^x(x+1) \\ &= e^x(x+2)\end{aligned}$$

Stationary points occur when $\frac{dy}{dx} = 0$
($e^x > 0$)

$$\therefore x = -2 \quad y = -e^{-2}$$

Test for type of stationary point. Find second derivative:

$$\begin{aligned}u &= e^x & v &= x+2 \\ u' &= e^x & v' &= 1\end{aligned}$$

$$\frac{d^2y}{dx^2} = e^x + e^x(x+2) = e^x(x+3)$$

When $x = -2$,

$$\frac{d^2y}{dx^2} = e^{-2}(-2+3) > 0$$

Hence $(-2, -e^{-2})$ is a local minimum.

(b) (3 marks) Point(s) of inflexion occur when

$$\frac{d^2y}{dx^2} = 0:$$

$$\therefore x = -3 \quad y = e^{-3}(-3+1) = -2e^{-3}$$

Testing for sign change:

- $x = -4, \frac{d^2y}{dx^2} = e^{-4}(-4+1) < 0$
- $x = -2, \frac{d^2y}{dx^2} > 0$

Hence $(-3, -2e^{-3})$ is a point of inflexion.

(c) (2 marks)

- $x = 0$

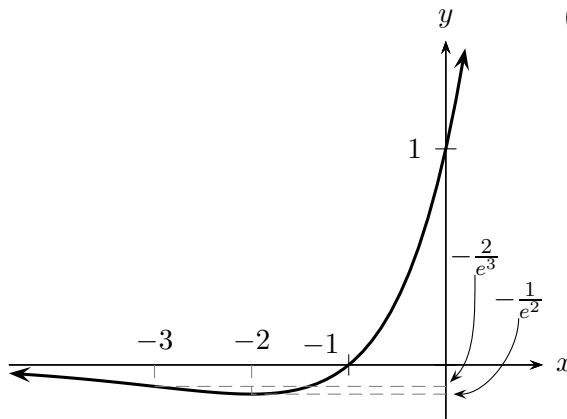
$$y = e^0(0+1) = 1$$

- $y = 0,$

$$e^x(x+1) = 0$$

$$\therefore x = -1$$

(d) (2 marks)

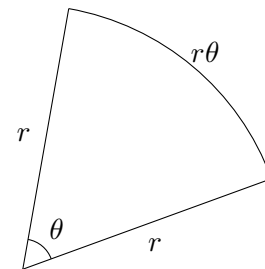


Examine $P(k+1)$:

$$\begin{aligned} & 4 \times 2^{k+1} + 3^{3(k+1)} \\ &= (4 \times 2^k) \times 2^1 + 3^{3k} \times 3^3 \\ &= (5N - 3^{3k}) \times 2 + 3^{3k} \times 3^3 \\ &= 5 \times 2N - 2 \times 3^{3k} + 27 \times 3^{3k} \\ &= 5 \times 2N + 25 \times 3^{3k} \\ &= 5(2N + 5 \times 3^{3k}) = 5P \\ &\text{where } P \in \mathbb{Z}^+ \end{aligned}$$

$\therefore P(k+1)$ is also true, and $P(n)$ is true by induction.

(b) i. (1 mark)



Question 9 (Lam)

(a) (3 marks) Let $P(n)$ be the proposition

$$P(n) : 4 \times 2^n + 3^{3n} = 5M$$

where $M \in \mathbb{Z}^+$.

- Base case: $P(1)$:

$$4 \times 2 + 3^3 = 8 + 27 = 35$$

Hence $P(1)$ is true.

- Inductive step: assume $P(k)$ is true for some $k \in \mathbb{Z}^+$, i.e.

$$P(k) : 4 \times 2^k + 3^{3k} = 5N$$

$$(N \in \mathbb{Z}^+)$$

$$\begin{aligned} A &= 32 = \frac{1}{2}r^2\theta \\ \therefore r^2 &= \frac{64}{\theta} \\ r &= \frac{8}{\sqrt{\theta}} \end{aligned}$$

ii. (2 marks)

$$\begin{aligned} P &= 2r + r\theta = r(2 + \theta) \\ &= \left(\frac{8}{\sqrt{\theta}}\right)(2 + \theta) \end{aligned}$$

iii. (3 marks)

$$P = 8\theta^{-\frac{1}{2}}(2 + \theta)$$

Differentiating,

$$\left| \begin{array}{l} u = 8\theta^{-\frac{1}{2}} \quad v = 2 + \theta \\ u' = -\frac{1}{2} \times 8\theta^{-\frac{3}{2}} \quad v' = 1 \\ \quad = -4\theta^{-\frac{3}{2}} \end{array} \right.$$

$$\begin{aligned} \frac{dP}{d\theta} &= 8\theta^{-\frac{1}{2}} - 4\theta^{-\frac{3}{2}}(2 + \theta) \\ &= 8\theta^{-\frac{1}{2}} - 8\theta^{-\frac{3}{2}} - 4\theta^{-\frac{1}{2}} \\ &= 4\theta^{-\frac{1}{2}} - 8\theta^{-\frac{3}{2}} \\ &= 4\theta^{-\frac{1}{2}}(1 - 2\theta^{-1}) \\ &= \frac{4}{\sqrt{\theta}}\left(1 - \frac{2}{\theta}\right) \end{aligned}$$

Stationary points occur when $\frac{dP}{d\theta} = 0$:

$$\begin{aligned} \frac{4}{\sqrt{\theta}}\left(1 - \frac{2}{\theta}\right) &= 0 \\ \left(1 - \frac{2}{\theta}\right) &= 0 \\ \therefore \frac{2}{\theta} &= 1 \\ \theta &= 2 \end{aligned}$$

Check second derivative:

$$\begin{aligned} \frac{d^2P}{d\theta^2} &= -\frac{1}{2} \times 4\theta^{-\frac{3}{2}} - \left(-\frac{3}{2}\right) \times 8\theta^{-\frac{5}{2}} \\ &= -2\theta^{-\frac{3}{2}} + 12\theta^{-\frac{5}{2}} \Big|_{\theta=2} \\ &= \frac{-2}{\sqrt{2^3}} + \frac{12}{\sqrt{2^5}} \approx 1.414 \\ &> 0 \end{aligned}$$

Hence $\theta = 2$ is a local minimum. P is minimised when $\theta = 2$.

$$P = \frac{8}{\sqrt{2}}(2 + 2) = \frac{32}{\sqrt{2}}$$